



North South University

Department of ECE

Handout: Database Normalization

(1NF, 2NF, and 3NF explained simply!)

Course: CSE311 – Database Systems

Introduction

Normalization is a process in **Database Design** used to organize data efficiently, reduce redundancy, and maintain consistency. It breaks large tables into smaller, related ones — while preserving data relationships using **foreign keys**.

Think of it like organizing your bookshelf:

- You don't keep the same book twice (no redundancy),
- Every shelf has a clear purpose (logical grouping),
- And everything is easy to find (consistency and clarity).

Normalization is achieved in steps called **Normal Forms (NFs)**.

1. First Normal Form (1NF)

Rule for 1NF

A relation is in **1NF** if *all attributes contain only atomic (indivisible) values*, and each record is unique.

Example:

StudentID	Courses
1	DBMS, AI
2	Networks

This table violates 1NF because the “Courses” field has multiple values.

Fix (convert to 1NF):

StudentID	Course
1	DBMS
1	AI
2	Networks

Now each field has atomic values — the table is in **1NF**.

2. Second Normal Form (2NF)

Rule for 2NF

A relation is in **2NF** if it is in 1NF *and* every non-key attribute is **fully functionally dependent** on the entire primary key.

Intuition: If a table's primary key has multiple columns, every non-key field should depend on *both*, not just one.

Example:

StudentID	CourseID	StudentName
1	DB101	Alice
2	AI201	Bob

Here, `StudentName` depends only on `StudentID`, not on both `StudentID` and `CourseID`. So, it violates 2NF.

Fix:

- Split into two tables:
 1. `Students(StudentID, StudentName)`
 2. `Enrollment(StudentID, CourseID)`

Now every non-key attribute depends fully on its table's primary key — the design is in **2NF**.

3. Third Normal Form (3NF)

Rule for 3NF

A relation is in **3NF** if it is in 2NF *and* there are no **transitive dependencies**.

Intuition: No non-key attribute should depend on another non-key attribute.

Example:

StudentID	DeptID	DeptName
1	D1	ECE
2	D2	CSE

Here:

$\text{StudentID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}$

`DeptName` depends on `DeptID`, which depends on `StudentID`. This is a transitive dependency.

Fix:

- Split into two tables:
 1. `Students(StudentID, DeptID)`
 2. `Departments(DeptID, DeptName)`

Now, `DeptName` depends only on `DeptID` — the schema is in **3NF**.

Summary Table

Normal Form	Key Idea
1NF	Atomic values (no repeating groups)
2NF	Full functional dependency on the primary key
3NF	No transitive dependency among non-key attributes

Final Tip

Normalization helps ensure your database is:

- Easier to maintain,
- Less redundant,
- More consistent and reliable.

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